

EXAM 7 – SPRING 2015

26. (2.5 points)

A primary insurance company is evaluating two reinsurance options. The company estimated the following information from its modeled probability distribution of underwriting results.

	<u>Gross</u>	<u>Option 1</u>	<u>Option 2</u>
Probability of Distress	3.00%	0.70%	0.65%
1-in-250 VaR Loss (\$000,000)	530	130	130
1-in-100 TVaR Loss (\$000,000)	570	220	210
Expected Net Underwriting Profit (\$000,000)	70	37	30

Cost of Risk Capital	10%
Required Risk Capital	1-in-100 TVaR

a. (1.5 points)

For each of the three risk measures provided, perform an efficient frontier analysis to determine whether either of the reinsurance options under consideration is superior to the other.

b. (1 point)

Using the cost of allocated risk capital method, determine which reinsurance option is preferable for the insurance company.

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EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT

QUESTION 26

TOTAL POINT VALUE: 2.5

LEARNING OBJECTIVE: C6: Describe the rationale for, methods for, and effect of managing insurance and financial risks.

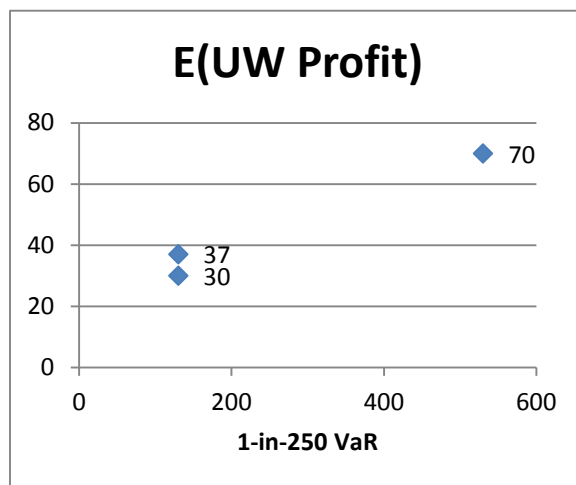
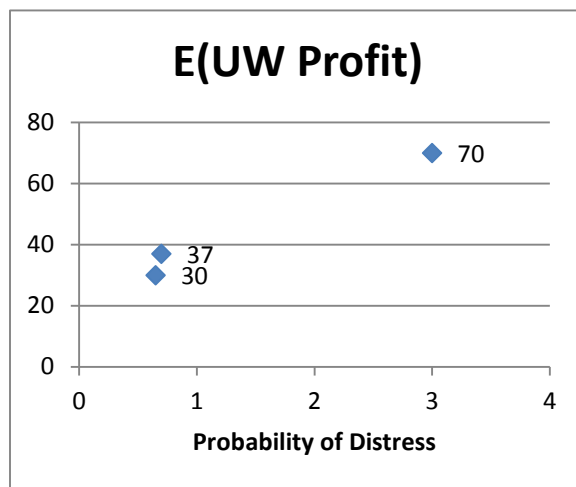
SAMPLE ANSWERS

Part a: 1.5 points

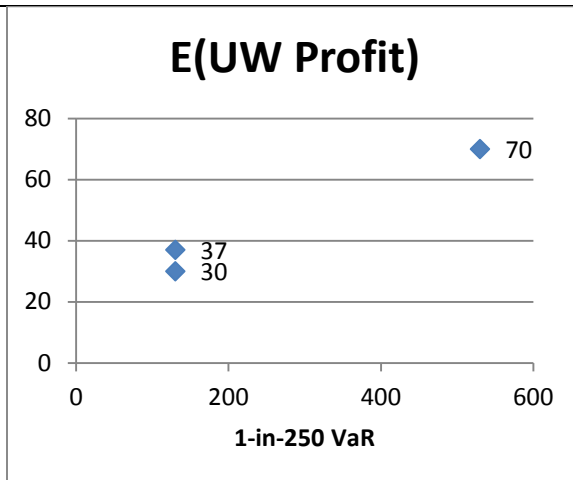
Sample Answer 1

Based on probability of distress and TVAR graphs, it's unclear which option is preferred, if any. Would be a management decision.

Looking at the VAR frontier, Option 1 is clearly better because it has a higher U/W profit for same amount of VAR.



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Sample Answer 2

(The same graphs were created as above. The candidate's conclusion sentences follow.)

Probability of Distress: No conclusion if either reinsurance option is better since option 1 has both higher risk measure and higher profit.

1-in-250 VaR: For same risk measure but with more profit, option 1 is better than option 2.

1-in-100 TVaR: No conclusion for same reason as probability of distress.

Part b: 1 point

Sample Answer 1

	Gross	Option 1	Option 2
Capital	570	220	210
Cost of Capital	57	22	21
Savings in Cost	0	35	36
Net Reinsurance Cost		33	40
		2	-4

Option 1 more preferable.

Sample Answer 2

	Gross	Option 1	Option 2
Reinsurance Cost	0	33	40
Required Capital	570	220	210
Cost of Capital	57	22	21
Total Cost	57	55	61

Option 1 is better since it has a lower cost.

Sample Answer 3

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		Gross	Option 1	Option 2
	Risk Capital	570	220	210
	Cost of Capital	57	22	21
	Expected Net Profit	70	37	30
	Adjusted Profit including Cost of Capital	13	15	9
Choose Option 1 – maximizes profit with cost of capital accounted for.				
EXAMINERS' REPORT				
Part a				
Candidates had difficulty with this question part. The most common mistake was to examine only one risk measure. Another less common mistake was to analyze all three risk measures, but only comment on one. Candidates were expected to be able to analyze all three risk measures and provide a corresponding analysis of the best option. To obtain full credit, a candidate needed to either graph all three risk measures and draw a conclusion based on all three graphs, or provide a detailed enough description and analysis of each risk measure that the graphs were not necessary.				
Part b				
Candidates generally scored well on this question, though several lost credit if they solely based their decision on either the cost of capital or net cost of reinsurance. To receive full credit, candidates needed to demonstrate the knowledge that both the cost of capital and cost of reinsurance need to be accounted for when determining which option is better.				